The most commonly used graphs are

(i) Histogram

(ii) Frequency Polygon

(iii) Frequency Curve

(iv) Cumulative Frequency Curves (Ogives)

4.4.1 Histogram

A histogram is an attached bar chart or graph displaying the distribution of a frequency distribution in visual form. Take classes along the *X*-axis and the frequencies along the *Y*-axis. Corresponding to each class interval, a vertical bar is drawn whose height is proportional to the class frequency.

Limitations:

We cannot construct a histogram for distribution with open-ended classes. The histogram is also quite misleading, if the distribution has unequal intervals.

Example 4.9

Draw the histogram for the 50 students in a class whose heights (in cms) are given below.

Height	111 – 120	121 – 130	131 – 140	141 – 150	151 – 160	161 – 170
Number of students	4	11	15	9	8	3

Find the range, whose height of students are maximum.

Solution:

Since we are displaying the distribution of Height and Number of students in visual form, the histogram is drawn.

Step 1 : Heights are marked along the *X*-axis and labeled as "Height(in cms)".

Step 2 : Number of students are marked along the *Y*-axis and labeled as "No. of students".

Step 3 : Corresponding to each Heights, a vertical attached bar is drawn whose height is proportional to the number of students.

The Histogram is presented in Fig 4.7.

For drawing a histogram, the frequency distribution should be continuous. If it is not continuous, then make it continuous as follows.